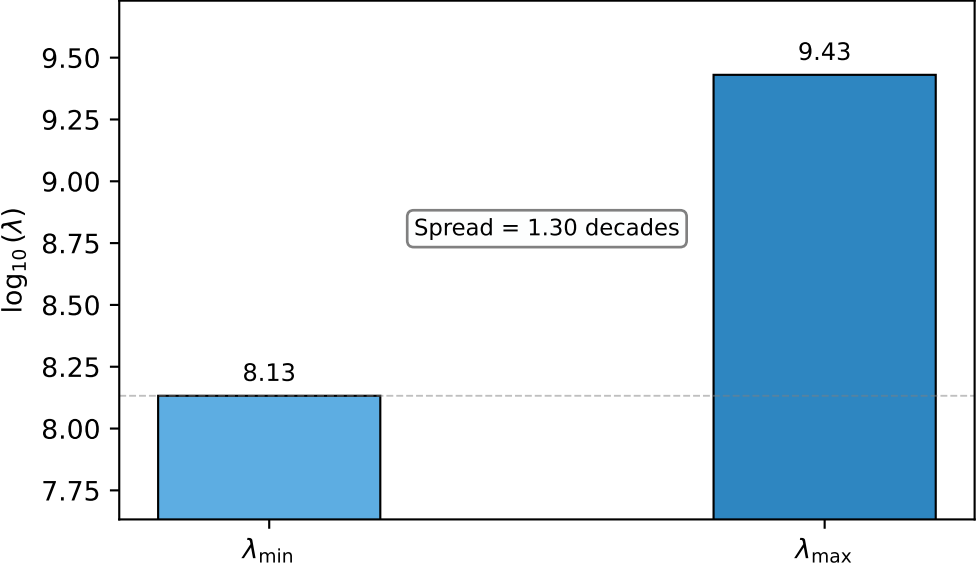


FIM identifiability analysis — 5-channel model
 $\kappa = 19.86$, eigenvalue spread = 1.30 decades

FIM eigenvalue spectrum



Identifiability audit verdicts

Metric	Value	Threshold	Pass?
Jacobian rank	2	2	✓
$\kappa(\text{FIM})$	19.86	< 1000	✓
Spread (decades)	1.30	< 3.0	✓
PL k_n two-sided	yes	—	✓
PL α_{tox} two-sided	yes*	—	✓

* α_{tox} CI hits lower grid boundary; profile flat on left (wide CI expected)
consistent with sub-EC50 linear regime.